Finance Data Analysis & Interactive Power BI Dashboard – Data Visualization Steps

1. Loaded the Cleaned finance dataset excel file in Power BI.
2. Cross-checked the table in Power query editor for errors and empty rows.
3. Checked the data type of each of the columns in all tables and made sure it was appropriate.
4. Closed and applied the changes done and loaded the table into Power BI.
5. Created the following measures for easy dashboard creation:
   - Average Transaction = AVERAGE(Cleaned_Finance_Data[Amount])
   - Total Transactions = COUNT(Cleaned_Finance_Data[Transaction_ID])
   - Total Transaction Volume = SUM(Cleaned_Finance_Data[Amount])
   - Successful Volume = CALCULATE(SUM(Cleaned_Finance_Data[Amount]),Cleaned_Finance_Data[Status]="Success")
   - Success Rate % = DIVIDE(COUNTROWS(FILTER(Cleaned_Finance_Data,Cleaned_Finance_Data[Status]="Success")),COUNTROWS(Cleaned_Finance_Data))
   - Failed Transaction Count = COUNTROWS(FILTER(Cleaned_Finance_Data,Cleaned_Finance_Data[Status]="Failed"))
   - Failed Transaction Volume = CALCULATE(SUM(Cleaned_Finance_Data[Amount]),Cleaned_Finance_Data[Status]="Failed")
   - Failure Rate % = DIVIDE(COUNTROWS(FILTER(Cleaned_Finance_Data,Cleaned_Finance_Data[Status]="Failed")),COUNTROWS(Cleaned_Finance_Data))
   - Fraud Transactions Count = COUNTROWS(FILTER(Cleaned_Finance_Data,Cleaned_Finance_Data[Fraud_Flag]="Yes"))
   - Fraud Percentage = DIVIDE([Fraud Transactions Count],[Total Transactions])
6. Also created the below hierarchy:
   - State Hierarchy: State →City →Bank Name→Account Type→Transaction Type→Payment Mode Updated→ Device Type
   - City Hierarchy: City →Payment Mode Updated→ Device Type→Platform
   - City Hierarchy 2: City → Bank Name→Account Type→Transaction Type
7. Added title using the text box in the Report view section of Power BI.
8. Created the following visualizations in the Report view section of Power BI.
9. In Report Page 1:
   - Total Transactions, Total Transaction Volume, Average Transaction were added using Card.
   - Success Rate, Failure Rate, Fraud Transaction Count were added using Multi-row card.
   - Count of Transactions by Status was depicted using a donut chart and a tooltip was also added to it.
   - Total Transaction Volume by Category was created using TreeMap.
   - Daily Transaction Trend was depicted using an area chart.

- Total Transaction Volume by State was created using Bar Chart. State Hierarchy was used to enable drill down functionality.
- Total Transactions vs Payment Mode was created using clustered column chart and a tooltip was also added to it.
- Slicers for Date and Payment Mode were added.
- Also, buttons for Drill Through and Navigation to Next Page were added.

10. In Report Page 2,
    - Failed Transaction Count, Failed Transaction Volume, Failure Rate, Fraud Transactions Count, Fraud Percentage, Successful Volume were added using Card visuals.
    - Failed Transaction Volume and Fraud Transactions Count by State and Fraud_Flag was depicted using Line and Clustered Column Chart. Drill through functionality was also enabled.
    - Total Transactions by City was showed using a funnel chart and a tooltip was also added to it.
    - Slicers for Date, Category and Fraud Flag were added.
    - Also, buttons for Drill Through and Navigation to Next Page and Back page were added.
11. In Report Page 3,
    - Table was added with the columns like Transaction ID, Date, Customer Name, State, City, Amount, Payment Mode, Status, Fraud Flag, Category, Bank Name, Account Type, Transaction Type and Device Type. Icons were added to Status and Fraud Flag. Data bars were added to Amount. It was the detailed table displayed when drill through was used.
    - Total Transaction Volume by City was displayed using Map Visual.
    - Slicers for Transaction Status (Failed or Success), Category and Fraud Flag were added.
12. Insights derived from the dashboard were added using Text box in Insights Page.
13. Recommended Business Decisions & Actions Based on Dashboard Insights were added in Business Decision Page.